

# Apratim Shukla

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## Education

### University of California, Davis

Master of Science in Computer Science, GPA: 3.88/4

September 2022 - June 2024

California, USA

### Vellore Institute of Technology

Bachelor of Technology in Electronics and Computer Engineering, CGPA: 9.28 / 10, Top 2%

July 2018 - June 2022

Chennai, India

**Coursework:** Algorithms, Databases, Software Engineering, Computer Organization and Architectures, Networks

## Experience

### ExpoLab, University of California, Davis (MERN stack, Flask, GraphQL)

December 2022 - Present

Graduate Student Researcher

Davis, USA

- Designed and optimized NexRes wallet, a Chrome extension enabling secure transaction submission and retrieval for web applications, resulting in a **10%** reduction in response time by implementing **Strawberry GraphQL** and **React**.
- Created a decentralized banking application for Ethereum, Radix, and NexRes, analyzing performance across all versions.
- Implemented and enhanced NFT marketplace using NexRes wallet, **Flask** backend, and **React** frontend, achieving a **12%** optimization through the implementation of **Python Asyncio** for various services.

### Metal Junction Services Limited (ELK Stack, Grafana)

August 2021 - February 2022

Data Analytics Intern

Kolkata, India

- Utilized **Logstash** for server events, logging data extraction and storing them in **Elasticsearch**.
- Visualized server CPU, memory, and I/O utilization using **Grafana**.
- Developed visualizations for logs analysis using **Kibana** and made predictions to improve system performance by **15%**.

### Center for Advanced Data Science, VIT Chennai (Python, Electron, Docker)

June 2021 - July 2021

Research Intern

Chennai, India

- Configured **Docker** on an **EC2** instance and deployed the required **MySQL** database.
- Upgraded the DSS for analyzing soil chromatography samples by incorporating new features like JSON-based interactive visualization in the **Electron** app.
- Implemented **Python** Scripting for automated extraction, manipulation, and uploading of data regarding soil chromatography and speeded up the process by **20%**.

### Poompuhar Shipping Corporation (MERN stack, GraphQL)

February 2021 - May 2021

Full-stack Intern

Chennai, India

- Built a ferry booking web application for searching ferries, booking rides, payment, and ticket generation.
- Collaborated with an intern to develop the admin dashboard and set up an API server with JWT-based authentication for mobile app development.
- Deployed the web application, admin dashboard, and API server on a **Digital Ocean VPS** and configured Cloudflare for DNS, SSL, caching, and protection.

### Fundboon Consulting Solutions Pvt. Ltd. (MERN stack, GraphQL)

May 2020 - January 2021

Full-stack Intern

Chennai, India

- Developed an admin dashboard with functionalities for Loan Applications, Bank Management, Lead Management, Profile Management, and User Management.
- Upgraded the user web application according to the specifications, implemented new features like OTP authentication, and loan application tracking, and optimized the request-response times by **50%**.
- Configured CI/CD for the application using **GitHub Actions** and deployed the web app on **Heroku**.

## Projects

### VITask | Web Scraping | [GitHub](#)

June 2019

- Stack/Tools:** Flask, Firebase, Electron, React Native, Flutter
- Founded VITask, a service that handles **400+** users and **100,000+** monthly requests. The project utilizes python libraries such as **BeautifulSoup** and **Requests** to fetch and parse the VIT Chennai VTOP website.

## Technical Skills

**Languages:** Python, C, C++, JavaScript, PHP, SQL, Verilog HDL, R, Java, Rust, Go, MATLAB

**Frameworks/Libraries:** BeautifulSoup, Requests, Socket.io, Tensorflow, Keras, Matplotlib, Numpy, Pandas, Scikit Learn

**Web Tech/Database/Tools:** HTML/CSS, Bootstrap, Flask, React, jQuery, MySQL, Docker, Git, AWS, GCP, Firebase, GraphQL